A fibre-wise variant of the n-connected postcomposition criterion

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'lem:nconnected_postcomp_variation' (pinned-original-source.tex lines 1265-1276, original Lemma 7.5.12); the proof environment opens at line 1278, one line after the blank separator, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 7, section 5, theorem offset 11) make the counted lemma print as Lemma 7.5.12, the number the first-edition label-number dump gives for 'lem:nconnected_postcomp_variation'.

Editorial note. The retained statement and proof use only macros of the book's own macro library and refer to no label outside the retained range; the only labels the proof uses are the two item labels declared inside the statement, so the package declares no external reference.

Editorial note. The retained preamble is the mechanical standalone adapter followed by the book's own macro library, reproduced verbatim from the same pinned source tree.

Editorial scope. One original result of the pinned first-edition chapter, counted once: the fibre-wise variant of the n-connected postcomposition criterion (author label lem:nconnected_postcomp_variation).

Allowed standard prerequisites (editorial).

- n-connected function. A function f:A->B is n-connected when the n-truncation of each fibre is contractible.. It is the property transferred between f and the map phi on total spaces.
- Fibre-wise n-connected family. A family g:prod_{a:A} P(a)->Q(f(a)) is fibre-wise n-connected when each g_a is n-connected.. It is the extra hypothesis on the family over f.
- Truncation of a fibre and contractibility. trunc n (hfib h y) is contractible exactly when h is n-connected at y, and contractibility is a mere proposition.. It is the form in which the equivalences of the proof are used, and the reason a merely inhabited fibre suffices.

Lemma 7.5.12. *Let $f : A \to B$ be a function and $P : A \to \mathcal{U}$ and $Q : B \to \mathcal{U}$ be type families. Suppose that $g : \prod_{(a:A)} P(a) \to Q(f(a))$ is a fiberwise n-connected*

family of functions, i.e. each function $g_a : P(a) \to Q(f(a))$ is n-connected. If f is also n-connected, then so is the function

$$\varphi : \Big(\sum_{a:A} P(a)\Big) \to \Big(\sum_{b:B} Q(b)\Big)$$
$$\varphi(a,u) :\equiv (f(a), g_a(u)).$$

Conversely, if φ and each g_a are n-connected, and moreover Q is fiberwise merely inhabited (i.e. we have $\|Q(b)\|$ for all $b : B$), then f is n-connected.

Proof. For any $b : B$ and $v : Q(b)$ we have

$$\begin{aligned}
\|\mathsf{fib}_\varphi((b,v))\|_n &\simeq \Big\| \sum_{(a:A)} \sum_{(u:P(a))} \sum_{(p:f(a)=b)} p_*(g_a(u)) = v \Big\|_n \\
&\simeq \Big\| \sum_{(w:\mathsf{fib}_f(b))} \sum_{(u:P(\mathsf{pr}_1(w)))} g_{\mathsf{pr}_1 w}(u) = \mathsf{pr}_2(w)^{-1}{}_*(v) \Big\|_n \\
&\simeq \Big\| \sum_{w:\mathsf{fib}_f(b)} \mathsf{fib}_{g(\mathsf{pr}_1 w)}(\mathsf{pr}_2(w)^{-1}{}_*(v)) \Big\|_n \\
&\simeq \Big\| \sum_{w:\mathsf{fib}_f(b)} \Big\| \mathsf{fib}_{g(\mathsf{pr}_1 w)}(\mathsf{pr}_2(w)^{-1}{}_*(v)) \Big\|_n \Big\|_n \\
&\simeq \|\mathsf{fib}_f(b)\|_n
\end{aligned}$$

where the transportations along $f(p)$ and $f(p)^{-1}$ are with respect to Q. Therefore, if either is contractible, so is the other.

In particular, if f is n-connected, then $\|\mathsf{fib}_f(b)\|_n$ is contractible for all $b : B$, and hence so is $\|\mathsf{fib}_\varphi((b,v))\|_n$ for all $(b,v) : \sum_{(b:B)} Q(b)$. On the other hand, if φ is n-connected, then $\|\mathsf{fib}_\varphi((b,v))\|_n$ is contractible for all (b,v), hence so is $\|\mathsf{fib}_f(b)\|_n$ for any $b : B$ such that there exists some $v : Q(b)$. Finally, since contractibility is a mere proposition, it suffices to merely have such a v. □